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Proposed experimental test of exotic monopole-dipole interaction at the micrometer range

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Here we propose an experiment to test the exotic spin-dependent interaction that can be mediated by axion-like particles. A sensitive cantilever is used to measure the force between a $Bi_4Ge_3O_{12}$ (BGO) microsphere and a spin-polarized magnetic structure. The magnetic structure consists of alternate reversely spin-polarized electrons. The magnetic structure is to be driven to oscillate during data acquisition, thus the exotic force signal is modulated to the high harmonics of the drive frequency. This helps to suppress the spurious background signal. The separation between the BGO sphere and the magnetic structure can be as close as 3.5 μm, and the force sensitivity is expected to be ~ 1 fN. If a Null result is obtained, the constraints on the coupling constant could be four orders of magnitude stronger than the current status of the experimental test at $\lambda = 10$ μm.

Keywords: Spin-dependent exotic interactions; axion-like particles; physics beyond standard model.

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1. Introduction

Theories describing physics beyond the standard model predict new particles and new interactions. The axion is one of them which has attracted strong attention, considering that the axion is a solution of the strong CP problem, and axion-like particles are candidates of cold dark matter.[1–6] The search for axion dark matter have been conducted through their coupling with photons,[7] such as ADMX,[8,9] CAST,[10,11] PVLAS,[12,13] and so on. Nuclei that are interacting with the background axion dark matter acquire time-varying CP-odd nuclear moments that can be detected by precision magnetometry.[14] Exotic spin-dependent interactions between two fermions can be mediated by axions through pseudoscalar coupling as first

proposed by Wilczek and his coworkers.[4] These interactions could be monopole-dipole interaction through scalar and pseudoscalar coupling or dipole-dipole interaction through pseudoscalar coupling only. Furthermore, spin-independent interaction could arise from two-axion exchange, and then the constraints on axion-nucleon coupling constants can be derived from the gravitational experiments or Casimir experiments.[15–19]

In this paper, we propose an experiment to test the monopole-dipole interaction between electrons and nucleons which is described by an effective potential

$$V_{md} = \frac{g_s^N g_p^e}{\hbar c} \frac{\hbar^2}{8\pi m_e} [\hat{\sigma} \cdot \hat{r}] (\frac{1}{\lambda r} + \frac{1}{r^2}) e^{-r/\lambda}, \tag{1}$$

where g_s^N and g_p^e are the scalar and pseudoscalar coupling constant, m_e is the mass of electron, $\hat{\sigma}$ is the spin unit vector, $\hat{r}$ is the unit position vector between electron and nucleon, and $\lambda = \hbar/m_b c$ is the interaction range with boson mass m_b. Such interaction can be equivalent to the effect of a spin in the exotic magnetic field $\vec{B}_{ex}$ produced by another fermion, i.e., $V(r) = \mu\hat{\sigma} \cdot \vec{B}_{ex}$. Using spin as a sensor, the interaction between different species of fermions can be tested by measuring the spin response to the exotic magnetic field, such as in experiments with magnetometers,[20–24] trapped ions,[25] nitrogen-vacancy (NV) center,[26] ultracold neutrons,[27] polarized ^{3}He atoms,[28–30] and dc SQUID.[31] On the other side, the spin-dependent macroscopic force can be measured by using torsion balance or micromechanical force sensor.[32–35] Except the experiment with NV center, most of them were performed at separations above a millimeter. Here we propose to search for such interaction by measuring the force between electrons and nucleons at the micrometer range using a soft cantilever.

2. Experimental Scheme

The experimental design is schematically shown in Fig. 1(a). A soft cantilever is used as a force sensor to measure the force between a $Bi_4Ge_3O_{12}$ (BGO) microsphere and a spin-polarized magnetic structure. We intend to search for the exotic interaction between the electron spins in the magnetic structure and the nucleons in the BGO sphere. The magnetic structure consists of magnetic blocks of alternative spin polarizations. As the potential is proportional to $(\hat{\sigma} \cdot \hat{r})$, the optimum spin direction should be along the z-axis in order to maximize the effect to be detected. However, as the spin polarization alternatively reverses in the z direction, the normal magnetic force is much larger than the designed force sensitivity of this experiment. In order to reduce the normal magnetic force, we add two magnetic shielding layers on the top and bottom surfaces of the magnetic structure, so that the magnetic field lines go from one magnetic block to another through the shielding layers. Since the electron spins are mostly in parallel with the surface in the shielding layers, their contribution to the exotic force are positive and

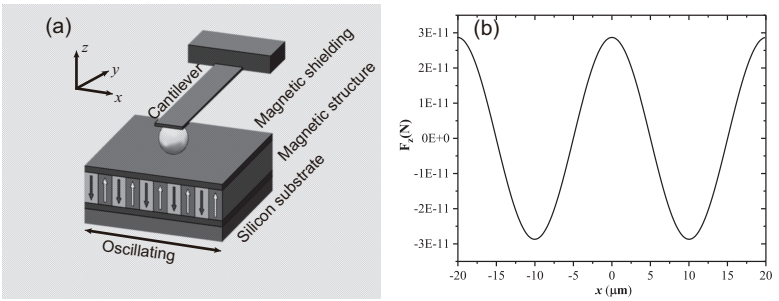


Fig. 1. (a) Schematic drawing of the proposed experiment. Dimensions are not in scale. The upside down arrows indicate the spin-polarization directions. (b) Calculation of the spin-dependent monopole-dipole interaction as a function of the lateral position x for $g_s^N g_p^e/\hbar c = 1 \times 10^{-14}$, $\lambda = 3$ μm.

negligible. During data acquisition, the magnetic structure oscillates along the x direction, thus the BGO sphere is subjected to a periodic varying force [Fig. 1(b)], and the signal is modulated to the high harmonics of the drive frequency. This helps to separate the signal of interest from other spurious signals in the frequency domain. Considering the limited travel range of the drive piezo, we plan to search for the exotic signal at the $6th$ harmonics of the drive frequency, and the drive amplitude is chosen to maximize the exotic effect to be detected. Similar modulation scheme has been used for the test of Yukawa type force in our previous experiment.[36]

3. Magnetic Structure

The preparation of the magnetic structure with controllable spin polarization is the main challenge of this experiment. We plan to fabricate such structure based on anodic alumina oxide (AAO) template which has ordered nano-porous structure [Fig. 2(a)]. AAO templates have been used to grow magnetic nanopillar arrays where every nanopillar is a single domain magnet due to the magnetic shape anisotropy.[37,38] The magnetization of the nanopillar can be manipulated individually with a magnetic force microscope (MFM).[39]

Micromagnetic simulation is used to calculate the hysteresis loop of single magnetic nanopillar of different materials and diameters using Mumax.[40] Based on the simulation, we choose to fabricate Nickel nanopillar arrays with diameter around 30 nm. We use 50 μm thick AAO templates with through nanopores. The average diameter of the pores is about 30 nm and the inter-pore distance is about 65 nm. The nanopores are filled with nickel by pulse electrodeposition.[41] In prior to electrodeposition, a 200 nm thick Au layer is deposited as working electrode on one side of the AAO membranes by thermal evaporation. To make the magnetization control feasible, we first grind the Ni filled AAO template to ~ 4 μm thick and then the surface is polished. Micromagnetic simulation shows that a single Ni nanopillar

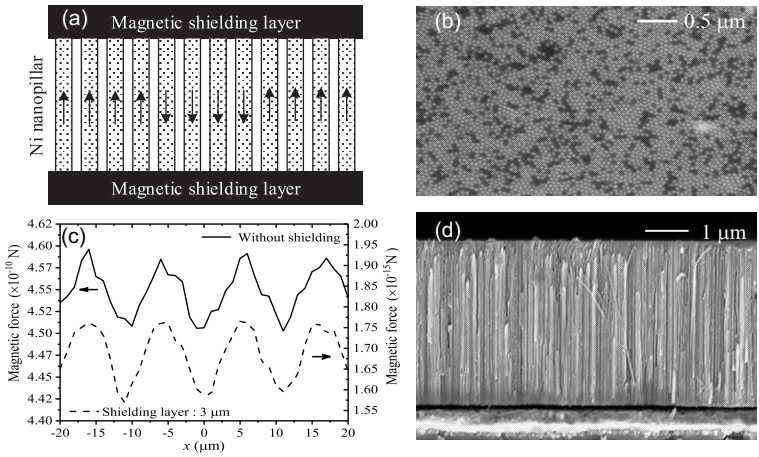


Fig. 2. (a) Schematic diagram of the magnetic structure. Nickel nanopillars are arranged in a hexagonal array with the magnetic shielding layers on the top and bottom surfaces. The arrows indicate different magnetization direction. Scanning electron microscopy (SEM) images of a nickel-filled AAO template after surface grinding, (b) top view, (d) cross-section. (c) Finite element simulation of the normal magnetic force acting on the BGO sphere when the shielding layers are included or not.

of 4 μm long has good squareness of the hysteresis loop with a cohesive field of 119 kA/m. Figures 2(b) and 2(d) show the top view and cross-section of one such magnetic structure after grinding and polishing. The average rms roughness after polishing is about 2.5 nm measured by an atomic force microscope.

The alternative anti-parallel magnetization will be prepared on a home-made MFM. The magnetic block width is optimized to be ~ 10 μm in order to maximize the signal at $6th$ harmonics with a limited travel range of the drive piezo. After magnetization writing using the MFM tip, a 3 μm thick permalloy film will be deposited either by electrodeposition or magnetron sputtering. We evaluate the normal magnetic force by finite element analysis with/without the shielding layers considered [Fig. 2(c)]. The simulation shows that 3 μm thick permalloy thin film can reduce the normal magnetic force by more than four orders of magnitude, thus the periodic variation of the normal magnetic force is suppressed to be less than 0.2 fN. The experimental separation could be as close as 3.5 μm by taking into account the thickness of the shielding layer and the surface to surface separation. Fig. 1(b) is the calculation of the exotic force as a function of the lateral position for this type of structure.

4. Force Sensitivity

We plan to use single crystalline silicon cantilever with spring constant of ~ 0.01 N/m. To have a high nucleon density, BGO microsphere with a diameter of 30 μm will be used. The microsphere is glued to the end of the cantilever and then

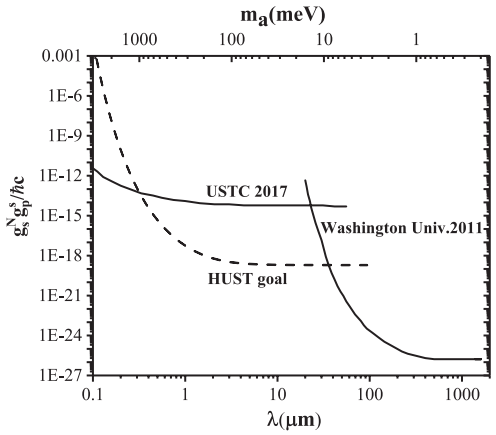


Fig. 3. Constraints on the exotic monopole-dipole interaction. The solid lines are results from Refs. 26 and 32. The dash line is the expected sensitivity of the proposed experiment. An upper bound of 1 fN is used to set the constraints.

coated with 100 nm thick Au. The thermal noise of the cantilever is estimated to be 1.7 fN/$\sqrt{\text{Hz}}$ assuming a quality factor of 6000.[42] With an integration time of 22 s as we used in the previous experiment,[36] the rms noise level is to be 0.4 fN. As the electrostatic force and Casimir force should not have the same periodicity as the magnetization modulation, their contribution could be negligibly small at the signal frequency. An optimistic force sensitivity could be $\sim$ 1 fN by taking into account the effects discussed above. If a Null result is obtained, we can obtain the constraints on the coupling constant using the force sensitivity as an upper bound of the exotic interaction. The projected sensitivity is presented in Fig. 3 with a upper bound of 1 fN. The expected constraints could be four orders of magnitude stronger than the current best result at $\lambda = 10$ μm.

5. Summary

In summary, we propose an experimental test of the monopole-dipole interaction which can be mediated by axion-like particles. We use a soft cantilever to measure the force between a BGO sphere and a magnetic structure with spin polarization modulation, and thus test the interaction between electron spins in the magnetic structure and the nucleons in BGO. The magnetic structure is composed of Ni nanopillar arrays fabricated on an AAO template, so that every nanopillar can be magnetized to up or down states using a MFM tip. The normal magnetic force variation is to be suppressed to $\sim$0.2 fN by the permalloy magnetic shielding layer, and the shortest separation could be 3.5 μm. Since the bulk effect with large amount of spins is measured, the constraints are expected to be four orders of magnitude stronger than the experiment using NV center quantum sensor at $\lambda = 10\mu$m.

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References

1. R. D. Peccei and H. R. Quinn, *Phys. Rev. Lett.* **38**, 1440 (1977).
2. S. Weinberg, *Phys. Rev. Lett.* **40**, 223 (1978).
3. F. Wilczek, *Phys. Rev. Lett.* **40**, 279 (1978).
4. J. E. Moody and W. Frank, *Phys. Rev. D* **30**, 130 (1984).
5. Q. L. Yang, *Mod. Phys. Lett. A* **32**, 1740003 (2017).
6. G. Bertone, D. Hooper and J. Silk, *Phys. Rep.* **405**, 279 (2005).
7. P. Sikivie, *Phys. Rev. D* **32**, 2988 (1985).
8. S. J. Asztalos *et al.*, *Phys. Rev. Lett.* **104**, 041301 (2010).
9. N. Du *et al.*, *Phys. Rev. Lett.* **120**, 151301 (2018).
10. C. E. Aalseth *et al.*, *Nucl. Phys. B* **110**, 85 (2002).
11. S. Andriamonje *et al.*, *Nucl. Phys. B* **138**, 41 (2005).
12. A. Ringwald, *J. Phys.: Conf. Ser.* **39**, 197 (2006).
13. F. Della Valle *et al.*, arXiv:1410.4081.
14. D. Budker *et al.*, *Phys. Rev. X* **4**, 021030 (2014).
15. E. G. Adelberger, E. Fischbach, D. E. Krause and R. D. Newman, *Phys. Rev. D* **68**, 062002 (2003).
16. E. G. Adelberger *et al.*, *Phys. Rev. Lett.* **98**, 131104 (2007).
17. V. B. Bezerra, G. L. Klimchitskaya, V. M. Mostepanenko and C. Romero, *Phys. Rev. D* **90**, 055013 (2014).
18. G. L. Klimchitskaya and V. M. Mostepanenko, *Phys. Rev. D* **95**, 123013 (2017).
19. G. L. Klimchitskaya, *Eur. Phys. J. C* **77**, 315 (2017).
20. B. J. Venema *et al.*, *Phys. Rev. Lett.* **68**, 135 (1992).
21. A. N. Youdin *et al.*, *Phys. Rev. Lett.* **77**, 2170 (1996).
22. K. Tullney *et al.*, *Phys. Rev. Lett.* **111**, 100801 (2013).
23. M. Bulatowicz *et al.*, *Phys. Rev. Lett.* **111**, 102001 (2013).
24. J. Lee, A. Almasi and M. Romalis, *Phys. Rev. Lett.* **120**, 161801 (2018).
25. D. J. Wineland *et al.*, *Phys. Rev. Lett.* **67**, 1735 (1991).
26. X. Rong *et al.*, *Nat. Commun.* **9**, 739 (2018).
27. T. Jenke *et al.*, *Phys. Rev. Lett.* **112**, 151105 (2014).
28. A. K. Petukhov, G. Pignol, D. Jullien and K. H. Andersen, *Phys. Rev. Lett.* **105**, 170401 (2010).
29. C. B. Fu, T. R. Gentile and W. M. Snow, *Phys. Rev. D* **83**, 031504(R) (2011).
30. P.-H. Chu *et al.*, *Phys. Rev. D* **87**, 011105(R) (2013).
31. W.-T. Ni *et al.*, *Phys. Rev. Lett.* **82**, 2439 (1999).
32. S. A. Hoedl, F. Fleischer, E. G. Adelberger and B. R. Heckel, *Phys. Rev. Lett.* **106**, 041801 (2011).
33. W. A. Terrano, E. G. Adelberger, J. G. Lee and B. R. Heckel, *Phys. Rev. Lett.* **115**, 201801 (2015).
34. G. D. Hammond, C. C. Speake, C. Trenkel and A. P. Patón, *Phys. Rev. Lett.* **98**, 081101 (2007).
35. T. M. Leslie, E. Weisman, R. Khatiwada and J. C. Long, *Phys. Rev. D* **89**, 114022 (2014).

36. J. B. Wang *et al.*, *Phys. Rev. D* **94**, 122005 (2016).
37. K. Nielsch *et al.*, *Appl. Phys. Lett.* **79**, 1360 (2001).
38. V. Vega *et al.*, *Nanotechnology* **23**, 465709 (2012).
39. K. Nielsch *et al.*, *J. Magn. Magn. Mater.* **249**, 234 (2002).
40. A. Vansteenkiste, *AIP Advances* **4**, 107133 (2014).
41. K. Nielsch, F. Müller, A. P. Li and U. Gösele, *Adv. Mater.* **12**, 582 (2000).
42. R. Castillo-Garza, C.-C Chang, D. Yan and U. Mohideen, *J. Phys.: Conf. Ser.* **161**, 012005 (2009).